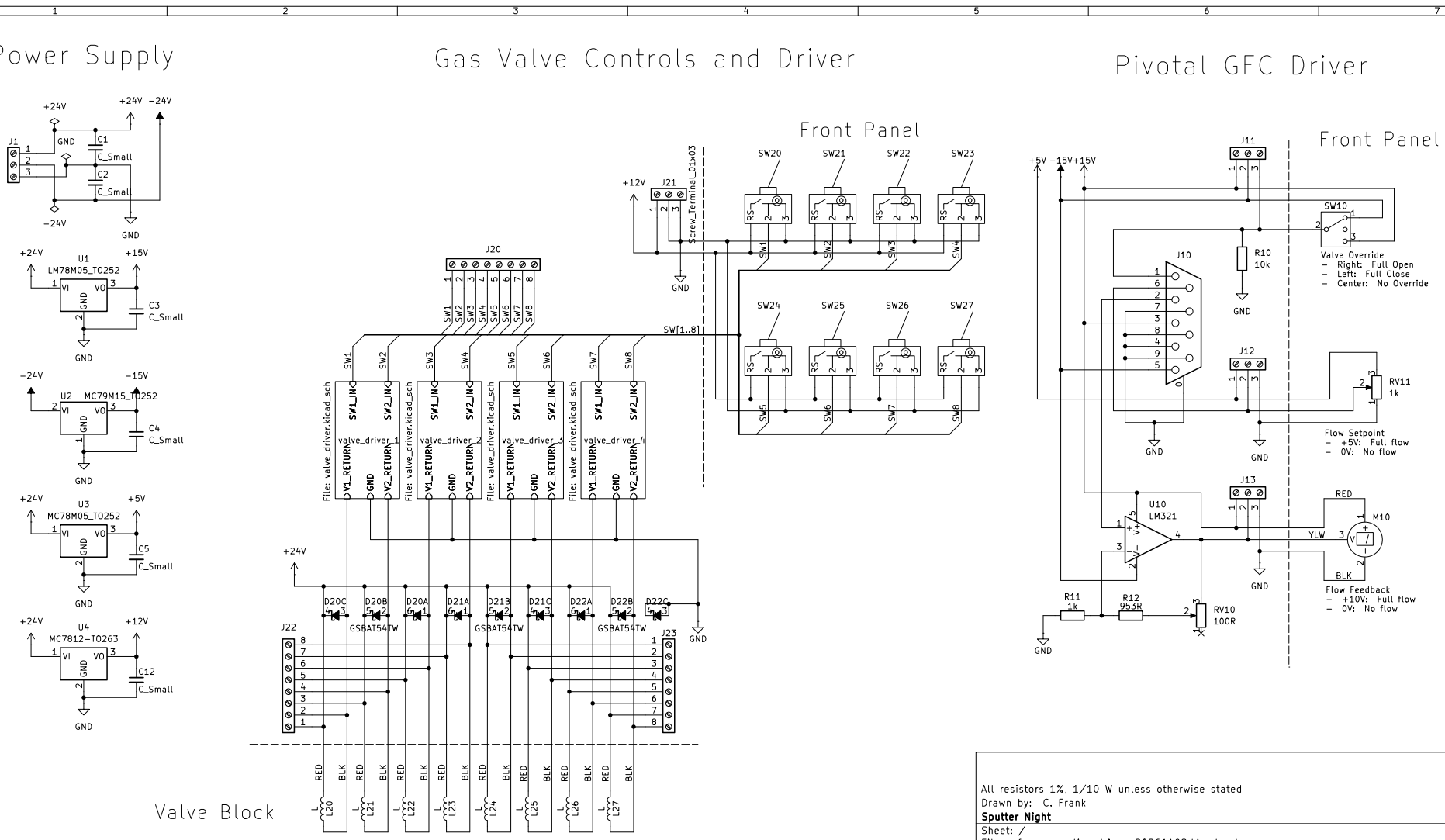
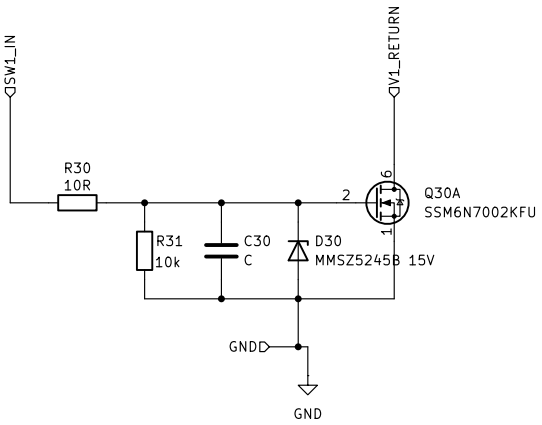
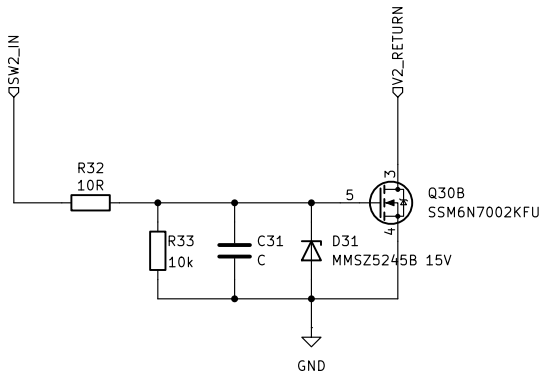

Power Supply
Gas Valve Controls and Driver
Pivotal GFC Driver
Front Panel
Front Panel
Valve Block
J1
C1
C_Small
C2
C_Small
+24V
+24V -24V
-24V
GND
U1
LM78M05_TO252
C3
C_Small
+15V
U2
MC79M15_TO252
C4
-15V
U3
MC78M05_TO252
C5
+5V
U4
MC7812-TO263
C12
+12V
J20
J21
Screw_Terminal_01x03
J22
J23
SW[1..8]
SW20
SW21
SW22
SW23
SW24
SW25
SW26
SW27
RS
valve_driver_1
valve_driver_2
valve_driver_3
valve_driver_4
File: valve_driver.kicad_sch
SW1_IN
SW2_IN
V1_RETURN
V2_RETURN
GND
D20C
D20B
D20A
D21A
D21B
D21C
D22A
D22B
D22C
GSBAT54TW
L20
L21
L22
L23
L24
L25
L26
L27
RED
BLK
J10
J11
J12
J13
SW10
Valve Override
- Right: Full Open
- Left: Full Close
- Center: No Override
R10
10k
RV11
1k
Flow Setpoint
- +5V: Full flow
- 0V: No flow
U10
LM321
M10
YLW
Flow Feedback
- +10V: Full flow
- 0V: No flow
R11
1k
R12
953R
RV10
100R
All resistors 1%, 1/10 W unless otherwise stated
Drawn by: C. Frank
Sputter Night
Sheet: /
File: mfc_pneumatics_driver-20261102.kicad_sch
Title: Gas Valve and MFC Driver
Size: USLegal
Date: 2026-11-03
Rev: 0.1
KiCad E.D.A. 8.0.9
Id: 1/5

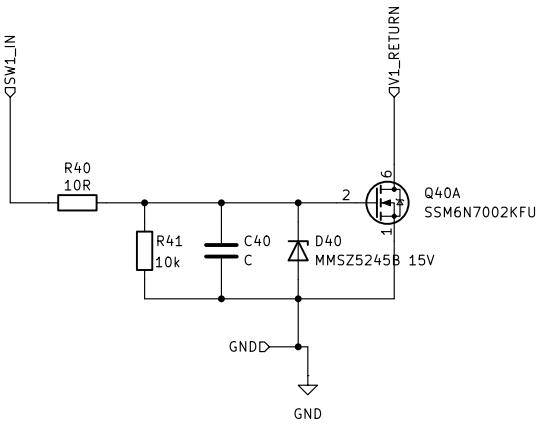


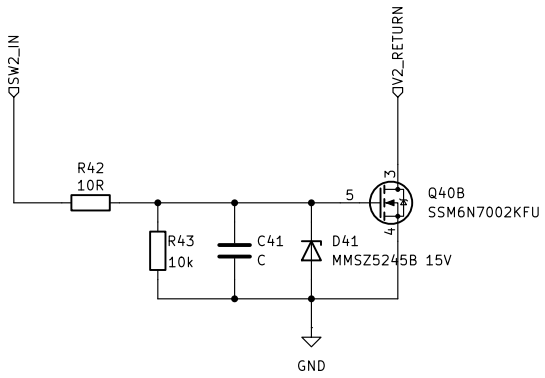


All resistors 1%, 1/10 W unless otherwise stated
Drawn by: C. Frank
Sputter Night
Sheet: /valve_driver_1/
File: valve_driver.kicad_sch

Title: Gas Valve Driver

Size: USLetter	Date: 2026-11-03	**Rev: 0.1**
KiCad E.D.A. 8.0.9		Id: 2/5



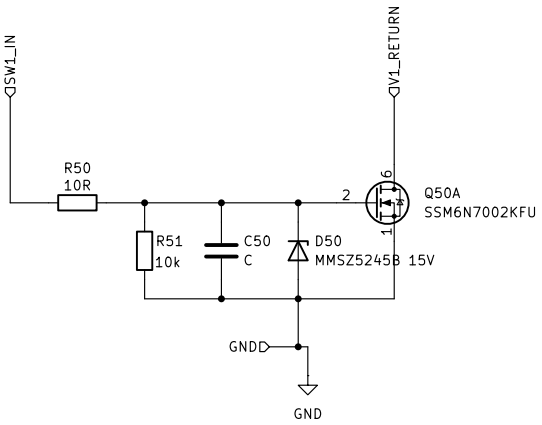


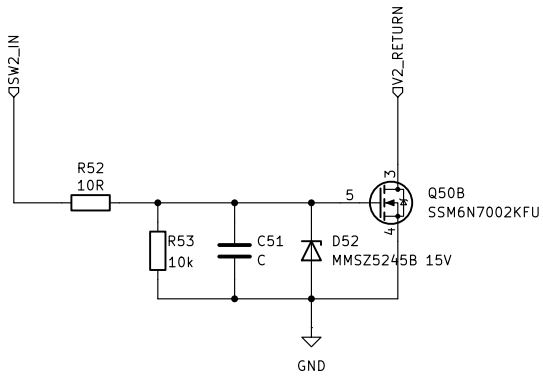
All resistors 1%, 1/10 W unless otherwise stated
Drawn by: C. Frank
Sputter Night

Sheet: /valve_driver_2/
File: valve_driver.kicad_sch

Title: Gas Valve Driver

Size: USLetter	Date: 2026-11-03	**Rev: 0.1**
KiCad E.D.A. 8.0.9		Id: 3/5



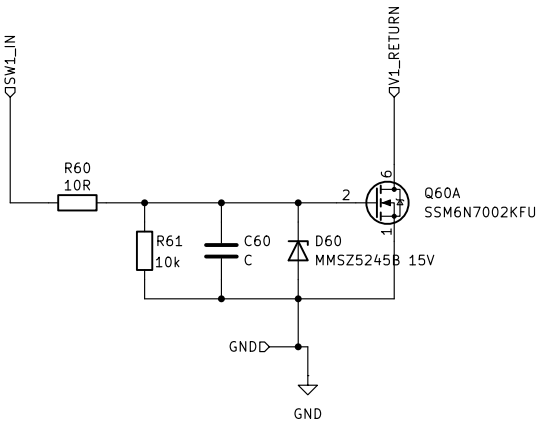


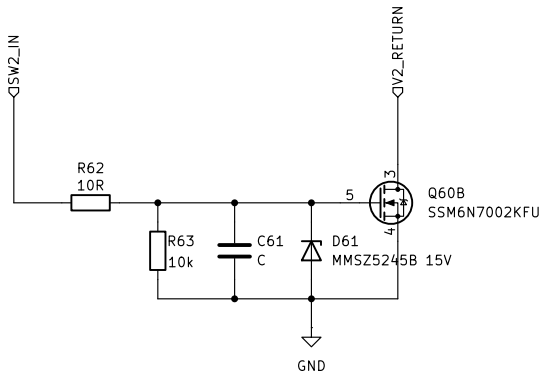
All resistors 1%, 1/10 W unless otherwise stated
Drawn by: C. Frank
Sputter Night

Sheet: /valve_driver_3/
File: valve_driver.kicad_sch

Title: Gas Valve Driver

Size: USLetter	Date: 2026-11-03	**Rev: 0.1**
KiCad E.D.A. 8.0.9		Id: 4/5





All resistors 1%, 1/10 W unless otherwise stated
Drawn by: C. Frank
Sputter Night

Sheet: /valve_driver_4/
File: valve_driver.kicad_sch

Title: Gas Valve Driver

Size: USLetter	Date: 2026-11-03	**Rev: 0.1**
KiCad E.D.A. 8.0.9		Id: 5/5